PSTAT 5LS | Lab 2

From guessing to simulation-based inference

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Today's plan

Today we will learn how to use **randomness and simulation** in R.

1. Generate random numbers and understand `set.seed()`.
2. Translate a research question into a chance model.
3. Use `simulate_chance_model()` to run simulations.
4. Visualize the results and estimate a p-value.

Work along in Lab 2 Notes

Open your Lab 2 Notes RMarkdown file. You will complete the code chunks and written answers during the lab.

1. Random numbers in R

Suppose we want to **randomly select five integers from 1 to 100**.

```
sample(1:100, size = 5)
```

- `1:100` is the set of possible integers.
- `size = 5` means select five numbers.

Live demonstration: Run the line twice. Do we get the same five numbers?

The question

If the output keeps changing, how could someone else reproduce our results?

2. Use `set.seed()` to reproduce results

Set the seed before generating random numbers:

```
set.seed(2573)
sample(1:100, size = 5)
```

Now run those same two lines again:

```
set.seed(2573)
sample(1:100, size = 5)
```

Both runs produce **exactly the same five numbers**.

- 2573 is an arbitrary integer, not a special value.
- `set.seed()` does **not** print a result by itself.
- Same seed + same random operation = same output.

3. What does setting a seed actually do?

A computer generates a **sequence of pseudo-random numbers**.

`set.seed(2573)` resets the generator to the **same starting point**.

Compare these two examples:

```
# Reset before each sample: matching results
set.seed(2573)
sample(1:100, 5)
set.seed(2573)
sample(1:100, 5)

# Set once: move forward in the sequence
set.seed(2573)
sample(1:100, 5)
sample(1:100, 5)
```

Important: Setting a seed does not make *every* random draw identical. It makes the **whole sequence reproducible**.

Your turn: Lab Notes

2. Our research question

Lee shows **40 students** pictures of **4 dogs**. Exactly one dog belongs to Lee.

Each student chooses the dog they think is Lee's.

17 of the 40 students guessed correctly.

If students are only guessing: each dog is equally likely to be chosen, so the chance of a correct guess is $1/4 = 0.25$.

What we want to know

Is 17 correct guesses out of 40 unusually high if students are just guessing?

3. Calculate the observed sample proportion

A **success** means a student selects the correct dog.

$$\hat{p} = \frac{\text{number of successes}}{\text{sample size}} = \frac{17}{40} = \boxed{0.425}.$$

Your turn: Sample Proportion

In the ‘sampProp‘ chunk, replace the placeholders and run:

```
phat <- 17 / 40
phat
```

Interpretation: 42.5% of students correctly identified Lee’s dog.

4. Write the hypotheses

Let p be the **true probability** that a student correctly identifies Lee's dog under this study's conditions.

$$H_0 : p = 0.25 \qquad \text{(no better than guessing)}$$
$$H_A : p > 0.25 \qquad \text{(better than guessing)}$$

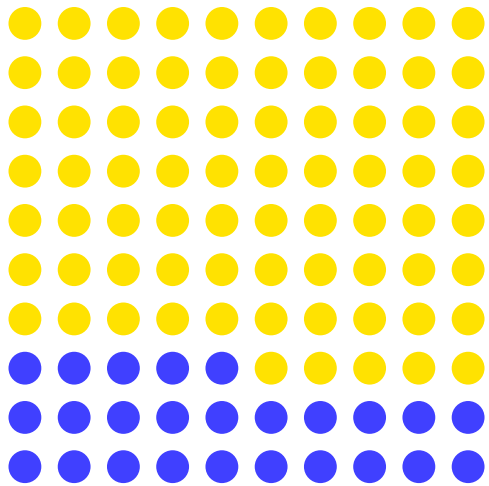
- The **null hypothesis** tells us how to simulate random guessing.
- The **alternative hypothesis** tells us which results count as more extreme: **larger** proportions.

Your turn: Hypotheses

Fill in H_0, H_A, and the meaning of p in your notes.

5. Build a chance model with poker chips

Imagine a bag with **100 chips**:



- **Blue chip (25%):** student guesses correctly.
- **Yellow chip (75%):** student guesses incorrectly.
- **One draw:** simulate one student's guess.

Draw, record the result, **replace the chip**, and mix again: each student's simulated guess has the same 25% chance.

6. What counts as one repetition?

One draw = one student's guess.

40 draws = one simulated class of 40 students.

For each simulated class, calculate:

$$\text{simulated } \hat{p} = \frac{\text{number of blue chips in 40 draws}}{40}.$$

One repetition produces **one simulated sample proportion**.

1,000 repetitions produce **1,000 simulated sample proportions**.

Your turn: Simulation Table

Complete the five rows in your notes: One draw, Blue chip, Yellow chip, Chance of blue, and One repetition.

7. Check your simulation table

Item	Meaning in our study
One draw	One student's guess
Blue chip	Correct guess
Yellow chip	Incorrect guess
Chance of blue	0.25 under H_0
One repetition	40 draws, one simulated class

Important distinction

40 draws are **within** one repetition. We repeat that entire 40-draw process **1,000 times**.

8. Translate the model into R

The function `simulate_chance_model()` has **three arguments**:

Argument	What it means	Here
`chanceSuccess`	Probability of success under H_0	`0.25`
`numDraws`	Students per simulated class	`40`
`numRepetitions`	Number of simulated classes	`1000`

Your turn: Explore the Help Page

Run the function help command in the aboutSimulate chunk. Find these three arguments in the help pane.

9. Run 1,000 simulated classes

Your turn: Code to Simulate

Complete and run the ‘runSim‘ chunk in your notes:

```
sim1 <- simulate_chance_model(
  chanceSuccess = 0.25,
  numDraws = 40,
  numRepetitions = 1000
)
```

- `sim1` stores **1,000 numbers**, one proportion per simulated class.
- You should see `sim1` in the Environment pane.
- If you run the simulation again **without resetting the seed**, the results will change.

Think: If guessing is random, should every simulated class get exactly 25% correct? **No.** Samples vary!

10. See the sample-to-sample variability

Your turn: Make a Histogram

Run the 'simHist' chunk in your notes:

```
hist(sim1,
     main = "Histogram of 1000 Simulation Results",
     xlab = "Simulated Proportion of Correct Guesses",
     xlim = c(0, 1))
abline(v = 17/40, col = "red")
```

- The **histogram** shows how simulated $\hat{p}$ values vary when H_0 is true.
- The **red vertical line** marks our actual result: 0.425.

Ask yourself: Is the red line near the middle or far into the right tail?

11. What does a p-value measure?

Our observed result is $\hat{p} = 0.425$.

If H_0 is true ($p = 0.25$), **how often would random guessing produce 0.425 or higher?**

$$\text{estimated p-value} = \frac{\text{simulated proportions } \geq 0.425}{\text{total simulated proportions}}.$$

Why **greater than or equal to**?

Because our alternative is $H_A : p > 0.25$: **larger** sample proportions are evidence in its direction.

Key idea

A small p-value means our observed result is unusual **if the null hypothesis is true**.

12. Calculate the estimated p-value

Your turn: Code to Find the Estimated p-value

Type this into ‘estpvalue‘ and run it:

```
sum(sim1 >= 17/40) / 1000
```

Read it in three steps:

1. `sim1 >= 17/40` checks which simulations are at least as high as our observation.
2. `sum(...)` **counts** how many meet that condition (`TRUE` counts as 1).
3. `/ 1000` converts that count to a proportion.

Your answer may differ slightly from the example p-value of about **0.006** because simulation involves randomness.

13. Make a conclusion in context

Compare your estimated p-value with the common significance level $\alpha = 0.05$.

- If **p-value** < 0.05, reject H_0.
- Here the estimated p-value is about **0.006**, so the result is unusual under random guessing.

Your turn: Conclusion

Write a complete sentence in the Conclusion section of your notes.

Model answer: We have statistically significant evidence that students can identify the teacher's dog **better than random guessing** in this setting.

Careful: This does not prove the explanation must be physical resemblance. Other cues could also matter.

14. Finish your Lab Notes

Before leaving, make sure you have completed:

1. `setSeed` and `aboutSimulate`
2. `sampProp` and the hypotheses
3. The five-row simulation table
4. `runSim` and `simHist`
5. `estpvalue` and the written conclusion

Three things to remember

Seed = reproducible randomness.
Simulation = what we might observe if H_0 were true.
p-value = how often results at least this extreme occur under H_0.

Questions? Use the remaining time to finish and check your notes.